

[Answer any five from the following questions]

1. a. Illustrate the relationships among system & application program, operating system and computer hardware. (3)
- b. How does an open source development model work? "Linux is an open source operating system" – Justify the statement. (3)
- ★* c. Define dual-mode operation of an operating system. Show the transition from user to kernel mode in dual mode operation with an example. (3)
- d. Suppose you want to move the contents of an existing file to a new file. Explain the process with system call sequence. (3)
2. a. What is context switch? Show with pictorial representation of Context switching. Give the brief descriptions of multilevel-feedback queue. (4)
- b. Cooperating processes need inter-process communication. What are two models of inter-process communication? Describe. (4)
- c. What are the benefits of using multi-threading instead of multiple process? Differentiate between the user thread and kernel thread. (4)
3. a. Consider the scenario of the read and write problems of operating system, one program writes and the other reads, and read and write don't interfere with each other. Design a solution to the problem using Semaphores ensuring the following conditions: (5)
- Common*
- i. If there is a read operation inside the critical section, no other read or write inside the critical section.
- ii. If there is a write operation inside the critical section, no other read or write inside the critical section.
- b. Analyze the Peterson's solution to decide whether it provides all the necessary requirements of synchronization mechanisms or not. (4)
- c. Criticize the Lock Variable mechanism of synchronization. (3)
4. a. CPU scheduling decisions take place when a process changes its states. Identify the changes of states when CPU scheduling is necessary. Also, classify them as preemptive and non-preemptive scheduling. (3)
- b. Three process P1, P2 and P3 arrive at time zero. The total time spent by the process in the system is 10ms, 20ms, and 30ms respectively. They spent first 20% of their execution time in doing I/O and the rest 80% in CPU processing. What is the percentage utilization of CPU using FCFS scheduling algorithm? (3)
- c. Consider the set of 4 processes whose arrival time and burst time are given below- (6)

Process	Burst Time	Arrival Time
P1	6	2
P2	2	5
P3	8	1
P4	3	0
P5	4	4

If the CPU scheduling policy is Shortest Remaining Time First (SRTF), calculate the average waiting time and average turnaround time.

5. a. For the following snapshot, draw the gantt chart using EDF algorithm

Process	Period	Execution Time
P1	10	5
P2	20	6
P3	15	3

Find the total utilization of the system.

- b. Assume that we have following resources: 5 pendrives, 2 printers, 4 scanners & 3 disks. The resources are allocated as per the matrix below: (5)

Process	Allocation				Max			
	pendrive	printer	scanner	disk	pendrive	printer	scanner	disk
P1	2	0	1	1	1	1	0	0
P2	0	1	0	0	0	1	1	2
P3	1	0	1	1	2	1	0	0
P4	1	1	0	1	0	0	1	0

- i) Find the available number of instances in the system for each of the resources.

- ii) What is the content of the max matrix?

Is the system in a safe state? If yes, find the safe sequence.

- c. Illustrate safety algorithm and resource-request algorithm with appropriate flowcharts. (3)

6. a. Given five memory partitions of 100KB, 500KB, 200KB, 300KB, and 600KB. How would the first-fit, best-fit, and worst fit algorithms place processes of 212KB, 417KB, 112KB, and 426KB? Which algorithm makes the most efficient use of memory? (6)

- b. Considering the following segment table and page table, translate the following logical address to physical address: (2, 6), where page size=2. (6)

Segment Table		
Segment	Limit	Page Table Base
S0	10	1000
S1	14	1200
S2	12	1400

Page Table		
P0	1000	F3
P1	1001	F2
P2	1002	F1
P3	1003	F0

$$P.A_2 = (0 \times 2 + 0) = 0$$

7. a. Define Hit ratio. Assume that a 90% hit ratio in the TLB 80% of the time and 10 ns is needed to memory. Find the EAT. (4)

- b. Calculate the size of memory if its address consists of 22 bits and memory is addressable. (4)

- c. Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB then determine the approximate size of the page table? (4)

Course Title: Computer Network & Internet Technology
Time: 3 Hours

Course Code: ICT-3103
Marks: 60

[Answer any FIVE from the following questions]

1. a. It has been said that flow control and congestion control are equivalent. Is this true for the Internet's connection-oriented service? Are the objectives of flow control and congestion control the same? (3)
- b. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable? (3)
- c. i) How long does it take a packet of length 1,000 bytes to propagate over a link of distance 2,500 km, propagation speed 2.5×10^8 m/s, and transmission rate 2 Mbps? (4)
 ii) Does this delay depend on packet length? $P.D = \text{distance} \div \text{p. speed}$
 iii) Does this delay depend on transmission rate? $T.D = \text{packet size} / \text{link's bandwidth}$
- d. Suppose end system A wants to send a large file to end system B. At a very high level, describe how end system A creates packets from the file. When one of these packets arrives to a packet switch, what information in the packet does the switch use to determine the link onto which the packet is forwarded? Why is packet switching in the Internet analogous to driving from one city to another and asking directions along the way? (2)
2. a. Using a suitable block diagram illustrate the satellite network. Also explain each of the block. (5)
- b. Describe the relative advantages and disadvantages of (3)
 - (i) Terrestrial links
 - (ii) Satellite links
 - (iii) Optical fiber transmission
- c. List the major components of an optical communication system and briefly describe each of them. (4)
3. a. Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why? Explain how it works? (5)
- b. Suppose Alice, with a web-based email account (such as Hotmail or Gmail), sends a message to Bob, who accesses his mail from his mail server using POP3. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application-layer protocols that are used to move the message between the two hosts? (4)
- c. Describe the role of a DNS on a computer network with reference to its components. (3)
4. a. Consider a planet where everyone belongs to a family of six, every family lives in its own house, each house has a unique address, and each person in a given house has a unique name. Suppose this planet has a mail service that delivers letters from source house to destination house. The mail service requires that (1) the letter be in an envelope, and that (2) the address of the destination house (and nothing more) be clearly written on the envelope. Suppose each family has a delegate family member who collects and distributes letters for the (6)

sender family members. The letters do not necessarily provide any indication of the recipients of the letters.

- i) Describe a protocol that the delegates can use to deliver letters from a sending family member to a receiving family member.
 - ii) In your protocol, does the mail service ever have to open the envelope and examine the letter in order to provide its service?
- b. Suppose Client A initiates a Telnet session with Server S. At about the same time, Client B also initiates a Telnet session with Server S. Provide possible source and destination port numbers for (6)
- i) The segments sent from A to S.
 - ii) The segments sent from B to S.
 - iii) The segments sent from S to A.
 - iv) The segments sent from S to B.
 - v) If A and B are different hosts, is it possible that the source port number in the segments from A to S is the same as that from B to S?
 - vi) How about if they are the same host?

5. a. Suppose Alice and Bob are sending packets to each other over a computer network. Suppose Trudy positions herself in the network so that she can capture all the packets sent by Alice and send whatever she wants to Bob; she can also capture all the packets sent by Bob and send whatever she wants to Alice. List some of the malicious things Trudy can do from this position. (5)

- b. Why Cookie is being used with HTTP server? Explain how Cookie works. (5)

- c. (HTTP and FTP are both file transfer protocols and have many common characteristics; for example, they both run on top of TCP. However, the two application-layer protocols have some important differences. Mention those differences.) (2)

6. a. Describe why an application developer might choose to run an application over UDP rather than TCP. (4)

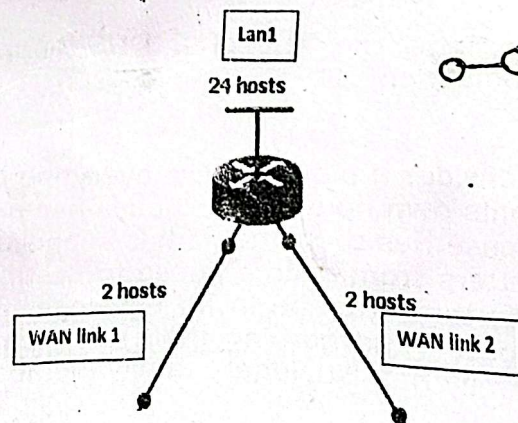
- b. (Suppose that a Web server runs in Host C on port 80. Suppose this Web server uses persistent connections, and is currently receiving requests from two different Hosts, A and B. Are all of the requests being sent through the same socket at Host C? If they are being passed through different sockets, do both of the sockets have port 80? Discuss and explain.) Common (4)

- c. Explain how Go-back-N protocol works. (4)

7. a. An ISP supplies a class B network of 136.210.0.0 to an enterprise that requires ten networks each to support 110 users. What is the network mask that would be configured in each workstation and what would be the NetID, Subnet Number? (4)

- b. Available IP Address: 192.168.2.0/24 (8)

Find the network ID and subnet mask for each network and show them in a table.





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Institute of Information Technology (IIT)

3rd Year 1st Semester B.S. (Hons.) Final Examination 2022

Course Title: Principles of Economics
Time. 3 Hours

Course Code: ICT-3109
Marks. 60

[Answer any five from the following questions]

1. a. A university produces two commodities: **research and teaching**. The resources the university uses include faculty and staff, libraries, classrooms, labs and so on. The following schedule indicates some points on the university's PPF: (12)

	A	B	C	D	E	F	G
Research	900	750	600	450	300	150	0
Teaching	0	20	45	75	110	150	200

- Define PPF. Does research production by the university exhibit increasing, constant or decreasing per unit opportunity costs?
- Define and graph the university's PPF and indicate efficient, inefficient and impossible points.
- Suppose the university is at point B but would like to alter production to point C. What would be the per-teaching unit opportunity cost of producing the extra teaching units?
- What will happen to the PPF if the main laboratory partially burns down? Graph the new PPF.
- What will happen to the PPF if all of the campus resources are doubled? Graph the new PPF.

2. a. What are the basic problems of micro economics? Explain. (3)

- b. What is demand? Why does the software demand curve downward sloping? Explain with graph. (3)

- c. Given a demand $Q_d = 100 - 5P$; where Q_d is the quantity demanded and P is the price. and a supply equation $Q_s = 80 + 5P$; where Q_s is the quantity supplied and P is the price, calculate the equilibrium price and quantity and show it in a figure. (6)

3. a. Draw and briefly explain the circular flow diagram. Identify the parts of the model that correspond to the flow of goods and services and the flow of taka for each of the following activities (8)

- Sorna pays a store keeper tk.200 for a quart of milk.
- Ontu earns tk.100 per hour working at a first food restaurant.
- Rimu spends tk.500 to see a drama at the theatre house in Green Road.
- Arif earns tk.10,000 from his 10 percent ownership of Acme Industrial.

- Distinguish between shift in demand and movement along the demand curve with figure. (4)

- a. What is supply? Draw the supply curve for a computer. If (I) technological innovation occurs and (II) raw materials prices increase in the economy, what will happen in the supply curve? Explain with relevant figures. (6)
- b. Differences between price elasticity of demand and price elasticity of supply? A company used to sell its product at Tk.120 per unit, the market demand was 50,000 units. When price was increased to Tk.190 per unit, its demand decreased 40,000 units. Calculate the price elasticity of demand and the effect of total revenue if demand is met in full at both the prices (6)

5. a. What do you mean by short run and long run phenomenon? Explain fixed cost, variable cost and total cost. (2)
- b. Differences between perfect competition market and monopoly market with examples. (3)
- c. Given the following total revenue (TR) and total cost (TC) functions for a firm, maximize profit (π) for the firm as follows: (7)

(i) Find the average revenue (AR), marginal revenue (MR), average cost (AC) and marginal cost (MC) functions.

(ii) Set up the profit function (π).

(iii) Find the critical value(s) where π is at a relative extremum and test second order condition and calculate the maximum profit.

$$TR = 1400Q - 7.5Q^2, \quad TC = Q^3 - 6Q^2 + 140Q + 750$$

$$\frac{TR}{Q}$$

$$TR - TC$$

6. a. The consumption bundles of the typical middle-class families of a city include 4 kgs of mangoes, 2 kgs of oranges, 2 kgs of apples, 1 kg of grapes. The market prices of these commodities are as follows: (base year 2001) (8)

Year	price of mangoes	price of oranges	price of apples	prices of grapes
2001	25	40	45	70
2002	29	45	52	85
2003	32	49	60	94
2004	35	56	64	100
2005	40	65	70	120

Calculate the consumer price index (CPI) and the inflation rate. What are the problems of calculating CPI?

- b. What does the GDP deflator and CPI measure? In what respects GDP deflator is different from the CPI. (4)

7. a. Define AD (aggregate demand) and AS (aggregate supply) with graph. Why does long run aggregate supply curve vertical? (8)

- b. Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve: (4)

- Bangladesh Govt. undertakes massive investment in physical and social infrastructure development projects.
- Net export level increases in the economy

- c. Distinguish between demand pull inflation and cost push inflation with graph. (4)